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- One-and two-qubit systems
- VQE
  - * Measurements in the computational basis
  - * Efficient calculation of gradients
  - * optimization

One qubit case

$$\mathcal{H} = a\mathbb{1} + bZ + cX + dY$$

ansatz $|0\rangle \longrightarrow U(\theta,\phi)|0\rangle$

$$= |\psi(\theta,\phi)\rangle$$

$$U(\theta,\phi) = R_x(\theta)R_y(\phi)$$

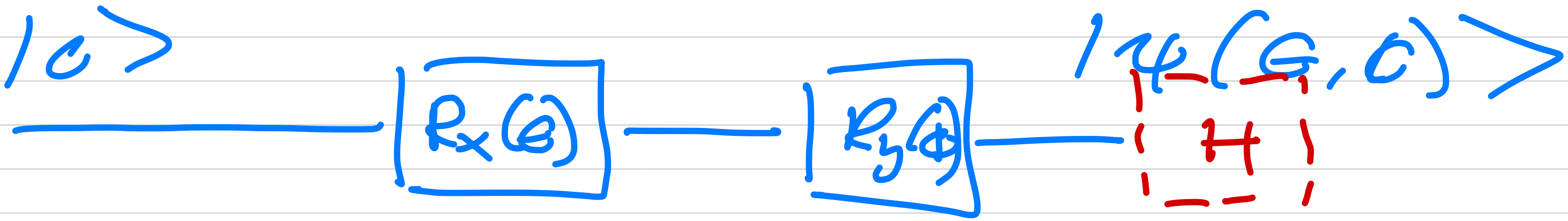
$$|\psi(\theta,\phi)\rangle = R_x(\theta)R_y(\phi)|0\rangle$$

$$|\psi\rangle_{exact} = \alpha|0\rangle + \beta|1\rangle$$

$$E[\phi] = \langle \psi(\phi) | \mathcal{H} | \psi(\phi) \rangle$$

$$\vec{\nabla}_\phi E[\phi]$$ parameter-shift rule

$$R_x, R_y \rightarrow R_i(\phi) = \exp\left(-i \frac{\phi}{2} \sigma_i\right)$$



$$\frac{\partial R_i(\phi)}{\partial \phi} = -\frac{i}{2} \sigma_i R_i(\phi)$$

$$\langle \psi(\phi) | \mathcal{H} | \psi(\phi) \rangle$$

$$= \langle \psi(\phi) | a \mathbb{1} | \psi(\phi) \rangle$$

$$+ \langle \psi(\phi) | b Z | \psi(\phi) \rangle$$

$$+ \langle \psi(\phi) | \overbrace{cX}^{cHZH} | \psi(\phi) \rangle$$

$$+ \langle \psi(\phi) | \overbrace{dY}^{S^{\dagger}HZHS} | \psi(\phi) \rangle$$

$|0\rangle$ — $\boxed{R_y(\phi)}$ — $\boxed{R_x(\theta)}$ — $\boxed{H}$ — $|\psi'\rangle$

$$R_i(\phi) = \cos\left(\frac{\phi}{2}\right)\mathbb{1}$$
$$- i \sin\left(\frac{\phi}{2}\right)\nabla_i$$

$$\frac{\partial}{\partial\phi}\langle\psi(\phi)|\mathcal{H}|\psi(\phi)\rangle$$

$$= \langle\psi(\phi)|\mathcal{H}\left(-\frac{i}{2}\nabla_i\right)|\psi(\phi)\rangle$$

$$+ h.c.$$

$$\langle \psi(\phi) | \mathcal{H} | \psi(\phi) \rangle = \langle 0 | R_i^{\dagger}(\phi)\, \mathcal{H}\, R_i(\phi) | 0 \rangle$$

$$= \langle \mathcal{H} \rangle$$

$$\frac{\partial \langle \mathcal{H} \rangle}{\partial \phi} = \langle \psi(\phi) | \mathcal{H} \left(-\frac{i}{2} \sigma_i \right) | \times | \underbrace{\psi(\phi)}_{R_i(\phi) | 0 \rangle} \rangle + h.c.,$$

$$\langle \psi | A^\dagger B C | \psi \rangle =$$

$$\frac{1}{2} \Big[\langle \psi | (A+C)^\dagger B (A+C) | \psi \rangle$$

$$- \langle \psi | (A-C)^\dagger B (A-C) | \psi \rangle \Big]$$

$$A = \mathbb{1} \qquad B = \mathcal{H}$$

$$C = -i \nabla_i / 2$$

$$\langle \psi | (\mathbb{I} - \frac{i}{2} \nabla_i)^\dagger \mathcal{H} (\mathbb{I} - \frac{i}{2} \nabla_i) | \psi \rangle$$

$$- \langle \psi | (\mathbb{1} + \frac{i \nabla_i}{2})^\dagger \mathcal{H} (\mathbb{1} + i \nabla_i / 2) | \psi \rangle$$

$$R_{n}(\phi) = \cos\left(\frac{\phi}{2}\right)\mathbb{1} - i\sin\left(\frac{\phi}{2}\right)\sigma_{n}$$

$$\phi = \pi/2$$

$$R_{n}(\pi/2) = \frac{1}{\sqrt{2}}\left(\mathbb{1} - \frac{i}{2}\sigma_{n}\right)$$

$$\langle\psi|\mathbb{1}^{+}\mathcal{H}\left(-\frac{i}{2}\sigma_{n}\right)|\psi\rangle =$$

$$\frac{1}{2}\Big[\langle\psi|R_{n}\left(\frac{\pi}{2}\right)^{+}\mathcal{H}R_{n}\left(\frac{\pi}{2}\right)|\psi\rangle$$

$$-\langle\psi|R_{n}\left(-\frac{\pi}{2}\right)^{+}\mathcal{H}R_{n}\left(-\frac{\pi}{2}\right)|\psi\rangle$$

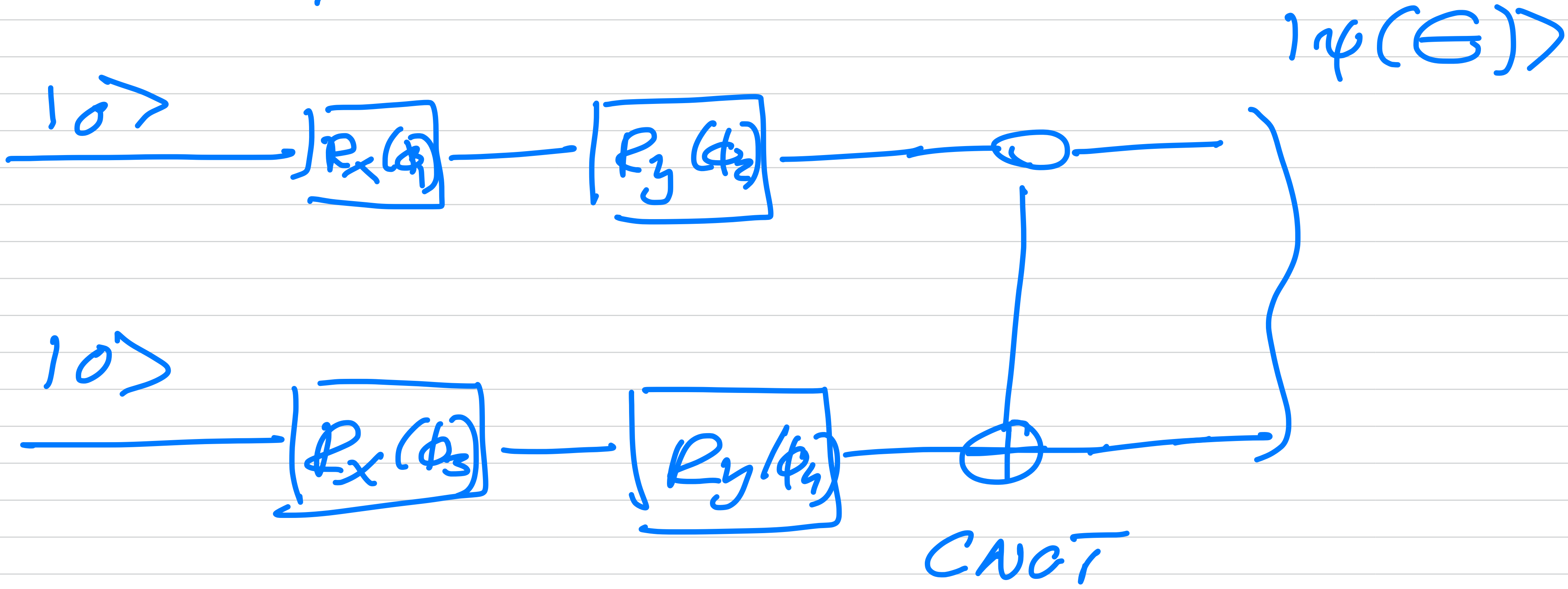
$$= \frac{1}{2}\left[\langle \mathcal{H}(\phi + \tfrac{\pi}{2}) \rangle - \langle \mathcal{H}(\phi - \tfrac{\pi}{2}) \rangle \right]$$

parameter-shift rule,

we will typically have many parameter

$$\Theta = \{ \phi_1, \phi_2, \phi_3 \ldots \}$$

Two-qubits



$$\vec{\nabla}_{\theta} \langle H(\theta) \rangle$$

$$\phi_i \leftarrow \phi_i - \alpha \vec{\nabla}_{\phi_i} \langle H(\theta) \rangle$$

learning rate

one qubit case

$$\langle \psi(\theta) | \mathbb{1} | \psi(\theta) \rangle$$

$$\{\phi_1, \phi_2\}$$

$$\langle \psi(\theta) | Z | \psi(\theta) \rangle$$

$$\langle \psi(\theta) \underbrace{H | Z | H}_{X} \psi(\theta) \rangle$$

$$\langle \psi(\theta) S^\dagger H | Z | H S \psi(\theta) \rangle$$